The mechanisms involved suggest an improvement in efficiency from
* Thermally insulating the outer pipe
* Placing reducing elements at the inner surface of the outer pipe, with high surface area if in the solid state, or as a liquid circulating blanket maintained by centrifugal forces in a rotating configuration
* Using spiraling vents at the entry of the exhaust into the cylindrical outer space, and coiled elements at the entry of the bubbled exhaust round the inner bar so that the flow spirals and, by the Ranque-Hilsch effect, concentrates its hot components on the outside and its cooler ones on the inside.
* Using a steel or alloy with high magnetic permeability and saturation, or very pure Iron for the bar and pipes
* Polarizing the fuel in the bubbler into a Magneliquid, and the fresh air into a Magnegas
* Increasing the pressure in the bubbler so that a maximal amount of carbon dioxide is dissolved
* Using a metallic powder in the bubbler so that the resulting carbonic acid releases hydrogen and carbonate in critical closed-cycle applications, or a multistage biomass of photosynthetic algae and anaerobic bacteria to convert the carbon dioxide into oxygen and biomass and the latter into methane in less critical or fixed applications.

You also mentioned reducing the rod length to reduce the icing problem.
How did you determine how much to reduce the rod length ?
There is some info in the GEET patent on that, (qv. files section for this list). My own tests are all cut and try at the moment. (No expert!) Rod length does not appear all that critical, but by decreasing the rod length, you can decrease the strength of the reaction and avoid the condensation issue. I think the first order purpose of the rod is to push the gases next to the wall of the chamber, where the heat is, and to create some resistance to the flow so that a partial vacuum is formed. The narrower the clearance, the faster the gases must flow, so the rod must get longer to allow enough time for the reaction to take place. Less clearance, less heating, slower movement, less peak vacuum.

As Paul mentioned here, the greater the temp difference between the vapors going into one end of the GEET, and the exhaust temp. going into the other end, the stronger the reaction will be. So too much preheating doesn't seem like a good idea.
****
And as for retrofitting F.I. vehicles... I just unplugged the computer in my Metro, wired in a vacuum advance ignition system, Installed the GEET, (a lot of design there) and that was about it...

**********_

It is simpler to tune the GEET for a small engine
than for a large engine (with variable vacuum, RPM, etc).
You can't use a rod from one reactor in another, if you do, it won't work in the first reactor anymore. Although you can run the exhaust back through the system, this gums up the carb eventually and reduces the temp differential. You don't require a rich fuel mix the reactor will work with any vapourized fuel mix from lean to rich and so on. To get the fittings real tight so they don't loosen under heat, preheat the fittings and tighten them hot. Ordinary pipe dope dissolves in gas, use automotive "form-a-gasket" that is made for heat and gasoline.

****

I changed the engine oil from synthetic to petroleum and all of a sudden there is no carbon buildup in the engine and exhaust like there used to be.

****

Make sure there are no vacuum leaks in the fuel intake line, from the engine intake, all the way through to the bubbler. Exhaust leaks should be sealed also but are not as crucial as having a good vacuum. Make sure that your bubbler is deep enough or has something in place to avoid liquid splashing into the fuel hose. Liquid will kill any reaction. The fuel vapor needs to be as "dry" as possible. Try different rod lengths. Longer rods are good for heavier fuels, shorter ones are good for lighter fuels. Perhaps try even a four inch rod. I have found iron to work as good as, if not better than, mild steel.
The clearances should all be according to what GEET suggests. Using small clearances increases the velocity of the gases, and thus increases the magnetic field that helps create the reaction in the chamber. "Burn in" your rod for at least 30 minutes, facing the part where the exhaust enters the chamber towards magnetic north. When you first cut the rod, it has a natural magnetic field on it. Make sure that you find the North end of the rod and place that in the chamber so that the North end is facing towards where
the exhaust is entering the chamber.

****

By increasing the vacuum efficiency of the unit, it will vaporize very easily at low temperatures, along with the petroleum-based fuels, without the need for applying external and or internal heat to the bubbler.

****

The fuel vapor must be as "dry" as possible.

****

I know that people have put coils on the chamber and pulled off enough current to fry their multimeter. And I have been told that pulling this energy off the chamber also kills the reaction.

Some researchers have found that the placement and length of the reactor rod was key to making the GEET fuel processor work or to perform at optimum efficiency.

****

My reactor is 6 inches or so from the exhaust manifold, so there is still enough heat for a reaction.

****

I have found that water droplets and gas droplets of any significant size kill the reaction almost immediately.

****

The GEET patent has much more info in it than the plans do.

****

You will hear a lot about vacuum leaks being hard to detect. Just use your unlit propane torch to move all around your piping joints any leaks will suck propane into the engine and change the engine speed.

****

My understanding of determining rod length is that you must position the rod so that it is facing east to west, and find where the flip is from north to south. Next you rotate the rod 180 and find the flip on the compass again. The space between the two flips should be the length of the rod.

****

I just tested my new GEET setup for fuel efficiency. It is now running 17.067 hours per gallon of gasoline. I still have quite a few tricks up my sleeve to get the efficiency way up.

****

Synthetic oil has been proven to ruin a reaction. It tends to coat and encapsulate as opposed to vaporize and atomize.

****

I noticed that when I got to the really high humidity that is over here in VA, I had to just about close the air intake all the way in order for the engine to run.

****

A vehicle bubbler should be durable (either metal or strong plastic) and be able to handle the fuel without dissolving (as some plastics do). PVC pipe or aluminum works well.

****

Use a float box to regulate bubbler fuel level.

****

There are certain tricks to upping the efficiency including making sure the fuel vapor is as "dry" as possible before it enters the chamber, using water injection, and using a temperature sensor to control the temperature of the fuel by switching between fresh air coming into the bubbler and hot exhaust.

****

To position the reactor rod, take the rod and put three little spot welds on each end, and then file them to the appropriate height to hold the rod in the center of the pipe. to hold it end to end, make a couple of twists of steel wire (non galvanized) and cut and bend them to hold the rod from sliding back and forth.

****
When you start the engine after the retrofit, you must have the exhaust end of the reactor chamber facing the magnetic north pole. Let it run for at least 30 minutes.
****

Based on a successful installation 7/22/00 of a GEET processor on a 3.5hp Techumsa rotary mower, some things we learned:

1. The rod must be secured with wire or twisted metal so it can't vibrate toward either end. We failed to do this at first and the engine started right up but then would not restart and it cost us several hours of frustration until we put a twisted wire inside the pipe to keep the rod from being sucked toward the engine. The rod had moved to one end and it tried to start but wouldn't keep running. At the end of the day, the rod vibrated to the other end and wouldn't run. A wire is needed at both ends!
2. Our setup is exactly as seen in the free plans. There are 4 valves in the setup. When we got it running properly, we had the air inlet valve closest to the engine (between the processor and the intake port) completely closed at all times. It wasn't really needed.
3. At first we had the about half of the exhaust diverted into the bubbler jar by closing down the valve between the processor tube and the exhaust muffler.
4. The throttle valve regulating how much vapor is allowed into the inner tube of the GEET Processor was about 3/4 open.
5. The air inlet valve at the TEE where the fuel vapor enters the inner tube of the processor tube was about 1/4 open. This mixed air with the fuel vapor prior to both entering the processor in a mixture that would allow the engine to run.
6. We spun the engine with a 1/2" Black & Decker electric drill. We cut off a socket extension rod, chucked that in the drill and put on a socket that fit the flywheel nut. We recommend you buy a regular nut from the hardware store to fit the crankshaft. The usual nuts are rather thin and a bigger one is better. We had to remove the rope starter mechanism. This works great to find the right valve settings. We tried a 3/8" drill but it would get very hot very quickly and was in danger of burning out. The 1/2"' drill will spin a 3.5hp all day and not sweat.
7. Once running, we found the throttle valve very sensitive (a 1/2" ball valve). It would go from idle to really flying in a very small valve movement. We tried brass gate valves also. They allow much finer adjustment but are slower to change and we found we liked the ball valves once we practiced awhile.

1. I had an elbow fitting right out of the exhaust manifold, which I replaced with a straight pipe. This had the effect of transferring more exhaust heat to the reaction chamber. (Its a little over 500' F. now)
I fear it is still too cold though. I may try some pipe wrap or insulation of some kind of muffler tape, etc.
2. I changed the configuration of the intake fuel vapor, and fresh air valves. I found that with the valves opposite each other as in the free plans, it would reach a break over point when the air valve was open too much farther than the fuel valve. This would cause the engine to stall easily when adjusting the fuel mixture. So I added a tee and routed it so the incoming fresh air pulled the fuel mixture out of its valve as it went by. This dumb little change actually helped quite a bit. In case the above didn't make sense.

```
air/fuel mix   <_________< fresh air valve
                          |
                          |
                          ^ fuel vapor valve
```

3. I added a throttle valve. While this is obviously not earth shattering, it did give me much more control over things. You can dial in the mixture and then adjust the throttle to obtain the desired rpm. (I got a little worried about over revs too)
4. All the tubes going to the bubbler are now metal. This transferred more exhaust heat to the bubbler and improved things a little. The bubbler is my current focus, the problem I am having is that as the engine runs the exhaust adds water to the fuel and the volatility goes down and down and down. Which means you have to adjust the fuel mix quite often. Its kind of amusing since the level in the bubbler doesn't change much, it just gets more watered down. It makes it look like it really isn't using any fuel at all. It is of course. It would seem to me you would have to have a bubbler temp over 200 degrees to vaporize the water as well. I am wondering if the water in the bubbler is really needed in an area that averages 90% humidity in the ambient air. Is anybody else running their Geet in a similar climate? (ie, when its not drizzle its rain) For those who have asked...Timing can be changed some by opening or closing the spark plug gap.
Wider=later. Or by changing the dwell angle, but most of the time the points are under the flywheel. Some also let you change the position of the magneto pickup. Any of these will affect engine timing. Hope that helps.
For the naysayers out there. First of all, build one of the devices before you go off commenting on the validity of it. My test engine HAS NO CARB

ATTACHED! And I am able to start and run the engine till the fuel supply runs out. It works, ok? But don't take anybodies word for it, build one yourself and see, like I did. Vapor carb devices have been around since the Otto cycle was new. There are over 600 patents on devices like this since 1880. Why is it that the last one in commercial production was in 1927 by the Holley carburetor Co.??? Ask an oil company, an automaker, or a politician. This is not Roswell stuff, most of the aspects of the geet technology are time proven science, and the plasma part is not understood by ANYONE yet. There is big money research going on at a lot of different universities and labs on just what makes a plasma field tick. So don't expect a bunch of garage hacks to explain it to ya. We can't just yet... Well..., Mr J.L. Naudin might be able to, he's one sharp cookie ;)

8. We could achieve very high rpm! Faster than I have ever heard one rev with a regular carb. I thought it would blow if we let it run that fast very long.

9. For the bubbler, we used a 2 qt. glass jar with a lid with fittings for exhaust to enter into a 1/2" copper pipe that had a cap soldered on the bottom and 4 holes about 1/16" dia drilled around its perimeter. This cap with holes on the side was about 1/4" from the bottom of the jug. We used about 2 cups of gas to start and that seemed fine. The other fitting sucked the fumes off the top into the processor.

10. The exhaust caused a white fog to form in the jar. The 4 holes produced plenty of agitation.

11. While it was running with exhaust recirculating into the glass jar, we removed the plastic hose feeding the exhaust into the jar and it kept running just as well just using the vacuum to draw the air through the 4 holes and create the agitation. The white fog disappeared when no exhaust was piped into the bubbler.

12. We mowed some high grass with it while I carried the jar alongside. We were not circulating the exhaust into the jar at this time. Had to go slowly because 3.5hp just isn't much power when in tall grass.

13. At one point we had a steady 6" of vacuum in the intake loop but this was with pretty watery gas in the jar. With fresh gas, it would only generate about 1/2 to 1" of vacuum.

14. With exhaust gas going into the jar, the fuel gets white and watery until it won't run very well. Without exhaust gas going into the jar, the fuel gets browner with time as you use the best stuff first. Fresh gas produces the most power.

15. One can use the plastic Prestone jug for a bubbler if you have the exhaust gas going into it. In this mode, the vacuum in the intake loop is

about 1/2" to 1" (pretty neutral) so the jug doesn't collapse. Without exhaust piped in, it collapses.
16. Without exhaust piped into the bubbler, the evaporation cools the fuel quite a bit and performance suffers. One really needs a way of utilizing the heat of the exhaust to warm it if you don't pipe it into the jar.
17. We found it better when we shortened the tubing to and from the bubbler. At first the tubes were about 3' long but this produced an oscillating motor speed and made it hard to keep running when we were just figuring it all out. When shortened to about 1 foot, response was faster.
18. We exchanged the blade for a steel disk 1/4" thick and 8" diameter. This worked great as an equivalent flywheel effect but much safer to work around. I highly recommend this for demo purposes. If your crankshaft is bent at all, the disk will wobble and the whole machine will hop around. Get another motor.
At the end of the day, we were able to throttle it down to a pretty low idle and back up to max rpm by adjusting both valves simultaneously. This took some practice. Butterfly valves linked together is the next step. We didn't do anything special to guide the gas or exhaust into the respective pipes of the processor. Just let it go in via the tees. The next step is to do this on a generator where power and fuel consumption figures can be compared. Hopefully by next week I have these.
****

... On the mower that Peter and I are working on, we have been finding a shorter rod (using gasoline as the fuel) gives an enormous increase in power, fuel efficiency, and emissions reduction.
****

" A few questions Greetings, > On the mower that Peter and I are working on, we > have been finding a shorter rod (using gasoline as the fuel) > gives an enormous increase in power, fuel efficiency, and > emissions reduction. This was basically what I found as well. In the patent, 4" is given as adequate for small engines, even though, on the phone, Paul said 7 1/4". This was a bit confusing. My initial test with the longer length gave very poor results. The rod was in the hot end of the reaction chamber, but carburetor kept getting too cold to work well. (Condensation) Over time the rod shook itself loose and eventually slid down to the cold end of the reaction chamber (my pipe slopes downhill), with only a few inches of its length actually in the zone of the exhaust pipe. The engine picked right up and was obviously running very well. I took things apart to see what was changing, and found the shifted rod
****

"The reactor rod, consisting of restrictive tubing, is connected to the inner chamber of the pyrolytic chamber at its one end and runs through the exhaust pie (centrally) towards the engine and exits approximately six inches

from the exhaust manifold where it connects with an air intake valve and from there enters the induction manifold. No modification to the engine itself is required if it is of the spark ignition type. Compression ignited engines may need some modification.
****
***Peter says: My friend Dan and I just tried using a water injection system using atomized water. It killed the reaction immediately. The fuel vapor must be as "dry" as possible. If I could inject a "drier" steam into the unit, I know it would work fine, providing the chamber was hot enough. My whole reason for trying that was to get the GEET to consume the water as fuel, since it did not seem to be vaporizing much into the chamber at such low temperatures, and another reason was to add "fresh" oxygen to the closed-loop system via the breakdown of the water. My next approach will be to try different valve arrangements to increase vacuum efficiency. As mentioned earlier, in a good vacuum, water will turn to steam at low temps.
****
† Bruce was ecstatic when I told him I got 82 MPG--however, Paul Pantone called me and was disappointed saying I should have gotten over 100 on that engine.† In any case, I built the reactor in the crossover pipe above the trans.† I didn't enlarge the pipe because I knew GEET gases have less volume than normally, with straight gas.† †† Next, I salvaged an old carb from the junkyard and removed the throttle plate from it complete with the linkages.† It was a bitch to remove the butterfly valve from the secondary barrel since the screws kept breaking and I had to drill them out.† Anyway, I had a guy weld up a piece of sheet metal with two 1 inch by 2" pieces of NPT pipe to fit in such a way that both pieces of pipe will center in their corresponding barrels.† Although it warped, I still managed to bolt it up to the throttle plate and seal it with silicone.† I drilled the holes to fit in the linkage shafts for both barrels and ground "custom" butterfly valves from the original aluminum ones.† The idea was to feed SNG into the primary (small) barrel and air into the secondary. This was part two complete. ††† Part three was the bubbler, a glass jar with a mickey mouse lid and an exhaust pipe surrounding the glass around the circumference (to heat the fuel).† Once the exhaust did the job of heating it ran inside to bubble through the gas.† I used a larger pipe to feed to the bubbler valve, an old lawnmower carb with all passages sealed up with silicone (only a temporary setup).† Then the other end went to the inlet side of the reactor.† The outlet of the reactor went straight to the primary barrel, which contained the butterfly valve I mentioned earlier.† As a first I just filled the bubbler jar manually.
††† The engine roared to life in an instant crank of the key.† However, I OVER estimated how much air the engine would need and discovered the air valve I machined didn't seal well enough.† So, I used a piece of the lid of

a pill bottle to block it off.† I'm assuming the engine sucked air in from beside the linkage shafts, because everything else was sealed.† I tried for acceleration with the pill bottle lid removed, of course, and found that some cylinders didn't receive an optimal mix (too lean).† This as I discovered was the fault of running the air and fuel into separate barrels. †† I corrected the problem by capping the secondary barrel and teeing both air and fuel into the primary. † I used the valve portion of an old lawnmower carb as the air valve, which I cut off the rest.† The adjustment was easy--just lengthen or shorten the linkage between the bubbler valve and air valve.† But still, the engine did not rev up too high, even though it was smooth.† As I let go of the throttle once, the engine bogged down to about 200 RPM or so, still Smooth but the oil pressure light started flickering. ††Yay.† Problem one solved. ††† I concluded that the bubbler was incapable of supplying enough fuel for the high rpm applications.† I switched to a primary carb instead--the one I used for a bubbler valve before with all the silicone removed.† Blah blah blah, going ahead...it ran rich all the time, no matter what.† And when I tried to lean it out it would stall ffffffasssst!!!† Then I tried removing the exhaust supply, since that is what was feeding the reactor--pure exhaust, no air.† Beauty--the engine ran great, but still chronic stalling, and a putrid richness factor.† At that point something hit me--I tried different positions with the fuel valve and it seemed useless in doing what it would be designed for.† I removed it and ran the engine again with the same adjustments as before...no more stalling.† I was able to lean it out so far, the smoke from the exhaust vanished.† A little further...oops--still stalling.† But problem two solved--the rich bitch factor. ††† With this solved I thought I'd give the bubbler another try.† First it was with exhaust.† Again, rich, but another dilemma arose--the mixture was unstable--sometimes rich, sometimes so lean the engine would stall.† At this point I used the starter so much I had to recharge the battery.† Okay, waiting period...after that, easy start.† I removed the supply of exhaust to the bubbler and found I was able to set up the mix better, more stable.† But still
another problem--no high revs.† I was able to take the engine up to 3500 RPM with the primary carb, only to 2400RPM with the bubbler, exhaust or air, doesn't matter.† with air however, the bubbler was ice cold--condensation formed on the outside of the jar...I'm surprised there wasn't any ice.† I finally came to the conclusion that the bubbler idea was garbage--besides, I tried a float system in it--do you have any concept of how hard it is to do something like that and make it work?† No, I didn't make it work. ††† Back to the primary carb...all set up, all as it should be.† I took the car out for a short trip but was not able to exceed about 80 km/h.† On my way

back I filled up, shutting off the engine.† Oh no!† Big mistake...it didn't want to start.† Plus, my starter was junk, so it didn't crank after a while.† Another waiting period, sitting at the pump, waiting for the engine to cool a little so the starter would work again.† I injected gas†into the air passage to aid in the restart.† Yay.† Whoopee.† I went home, and secured some of the mickey-mouse linkages to work more reliably.† NEXT DAY:† I went on my 100 km. test.† Filled up with 15 litres of gas, drove exactly 100 km. and came back, emptied the tank.† I can't recall the calculations off the top of my head, but in the end I converted km to miles and litres to gallons and came up with around 82 miles per gallon.† When I find that paper I'll list the calc's if anyone wants. †††† But there were problems.† After the 100 km. test which happened to be a Sunday, I drove the car the next day.† Cold starts were a horror--had to get out, pop the hood and inject gas into the air plenum to prime the reactor.† I had not set up a choke yet.† But I got fed up so I reverted to the carb, with which I can squeal my tires and start the car properly.† BUT it was a start--even though the linkages and cheapass welds fell apart on me I knew what I was in for. †††† Now I am onto design Two--the professional setup, the final setup, the flawless system.† Bruce was supposed to post some drawings of mine in the supercarbs files, I don't know what he did with them.† I want everyone to see them and for some reason he does not, and I can't put them in caus I have no scannner. †††† I hope this "life's story" will aid some in this...GEET stuff (well, maybe not--maybe I'm just wasting your time). Thank You

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> clearances should all be according to what GEET suggests. Using small
> clearances increases the velocity of the gases, and thus increases the
> magnetic field that helps create the reaction in the chamber.
****

Comments By Professor Shields on GEET

In response to many emails that I am receiving regarding the GEET technology, I shall explain AERO's standpoint. We never represented that the device is inoperable. In fact, we clearly state that the device DOES operate a standard 5hp gasoline engine. This device does replace a conventional carburetor. Our tests are continuing as you read this page. The points to be made as an UNBIASED, independent accounting of the operation of the device according to our CURRENT tests are as follows:

* The device appears to emit far less emissions than a standard carb.
* The device appears to exhibit unusual electromagnetic characteristics.
* The exhaust of the device is noticeably

* The device WILL operate on unconventional fuels.

Several points that do need to be made are as follows:

1. There is residue left over from the operation of the device.
2. The residue is in the volatilization chamber.
3. At this point we are unable to burn any noticeable amount of alternate fuel.

Anyone familiar with chemistry will soon realize that it is impossible to burn a solid without first having an intermediary gasseous compound. In gasoline there many different compounds. When the GEET device is in operation, it does act as a, "miniature refinery"; as Pantone states. The process is called fractional distillation. However, the vapor-point of the various compounds within the gasoline. The exhaust that is being bubbled into the volatilization chamber is so cool that, even under high vacuum, it is not enough to vaporize the solids.

****

I am using a modified carburetor to vaporize my fuel, where should I have the throttle setting? or am I supposed to use that to meter the fuel? I currently have another butterfly valve controlling fuel, and the modified carb's throttle held all the way open. I have a spring-loaded valve maintaining the necessary vacuum

****

I disagree. I've run my GEET mower with the exhaust line to the bubbler disconnected (just on a hunch), and it seems to work better. The reason I disconnected it? The large amount of water vapor leaving the exhaust was getting added into the gasoline in the bubbler, creating a yellow, odorous mixture. After a while, the mower would cease to run, although there was still plenty of fuel left.

So, I thought I'd let atmospheric pressure do the work. The vacuum from the fuel line exiting the bubbler was enough to draw air through the bubbler pipe. This enabled the mower to burn ALL of the gasoline in the tank, right down to the last bit of vapor. Just speculation here, but I suspect the engine is putting out almost, or as much, water vapor as it is burning fuel vapor. I suspect that letting the water in the exhaust mix with the gasoline is causing it to lose its volatility, and leave so much unused.

Also, letting fresh atmospheric air into the bubbler instead of exhaust requires setting the mixture valves a little differently, but it makes more sense. Internal combustion requires fresh air (with its oxygen).

Anyone else care to try it that way? Let me know what you discover.

****

I found a 3.5 hp briggs mower that would start but not run because it had a bad carb. Wasn't that just the right find for a geet carb?

1. left the stock carb for butterfly/govenor throttle control.
2. Used 1/2 gallon pickel jar bubbler as on previous 10 hp tecumseh.
3. Used only 1 air bleed ball valve near the carb to lean out the geet gas.
4. Started up nice and had almost full rpms. Throttled back too because of the stock carb butterfly control. If the jar was mounted it could have mowed grass.
5. Slowed and almost stopped after awhile.
6. Bubbler jar was rather cold so I put it in a bucket of hot water and picked up some rpms.
7. No obnoxious exhaust fumes in the garage as with the stock carb.
8. Started up without the rod and later inserted the rod. So it was running on vapor - not geet gas at first.

Next I want to try a pc carb to see if that makes any difference.
The problem with this engine is since I do not have a good stock carb to do original time tests I know nothing about geet excpet the exhaust runs clean.
****

I heated my bubbler jar with the exhaust up to 230 degrees and had complete evaporation and no residue or junk in the jar.
My muffler is mounted vertical and is 1.5 inches by 4 inches.
On top of that I attached a juice can 4.25 inches by 7 inches. With a hole in the bottom for the muffler exhaust to enter the can.
My bubbler jar is a 1-quart pickle jar with 6 1/16 holes around the tube.
The jar is dropped down into the can and heated with the exhaust.
PLEASE TRY this and you will be VERY PLEASED.
****

I just retrofitted a 10 HP genset here and it is running in closed-loop. Right now the exhaust is running completely into the bubbler, but I will soon be directing a portion of the exhaust into the intake as well. The power seemed excellent but we need to run some tests on it hooked up to some heaters to provide a good load. The noise was reduced by around 40-50% when the exhaust was funneled into the bottom of the bubbler. There were some places that still needed to be welded along the exhaust path and we expect this to reduce the noise level further. I was amazed at the reduction of noise we already attained.
****

I tell you what. I'll run you through it basically

1) Cheap 1/2" steel rod (12" long) in cheap 5/8" NPT pipe.
2) Cheap throttle portion of a lawnmower carb cut off of rest of carb, mounted as air valve, and a full lawnmower carb mounted as fuel valve and venturi
3) connect valves to appropriate locations with pipe (fuel valve to intake of reactor, air valve to a tee on intake manifold, output of reactor to other end of tee) and connect them with linkages.
4) Set the ratios and away u go!!!
That sounds simple and there are soooooooooo many problens i encountered.
****
It would run with mixtures such as 15% petrol / 85% water, Coke or urine. But it always needed some petrol. When the engine stopped it was hard to tell whether any of the non-petrol fuel did get consumed at all as a fair amount of liquid was still left in the bubbler. I have since st up a bubbler with a float valve, fed by a separate tank. I am using a second bubbler in series with the first one, which I fill with other fluids. On this setup, I managed to run the engine for several hours non stop as the fuel characteristics in the bubbler remain fairly constant now. Air inlet into the bubbler takes place via a constant vacuum valve of my own design. I am not using the exhaust feedback, as this does not seem to work very well on this engine. I am getting 150deg C from one end of the reactor chamber to the other. That's measured on the outside wall of the pipe which takes into account the thermal conductivity of the metal. The temperature drop of the gas should therefore be even greater. Exhaust temperature is around 32deg C which is just lukewarm
****

Most of the chambers that Peter and I have designed are made out of galvanized pipe nipples and fittings. We have used 1/2" x 12" for the inner chamber and 1" x (about) 7 1/2". Instead of having to drill a hole in both ends of the outer chamber for the inner chamber, we used 1" galv. tees with 1" x 1/2" bushings (which need to be threaded through with a 1/2" tap) to avoid excess machining. We assemble it by first attaching both tees to the ends of the outer chamber. Next, put one of the bushings on the end of the smaller pipe (inner chamber), and screw it into the end of a tee. Finally, you have to put the other bushing on the end of the other tee, while making sure that the inner chamber threads into the inside of the bushing. This part can be a little bit tricky, but if you have the right size parts, it will work. If you want to get the idea of the layout, go to a plumbing store and see if it will all

fit. If you cannot find the right size parts, you may have to have some pipe cut and threaded, but this seems to be the easiest overall design. Be sure to seal the threads when you are done!!! Also, rod lengths will vary from system to system and fuel to fuel. For gas, you will never need longer than a 7 1/2" rod, for alternate fuels, start with a 12" rod. Once it is burned in, you can determine what length is needed with the following procedure:
My understanding of determining rod length is that you must position the rod so that it is facing east to west, and find where the flip is from north to south. Next you rotate the rod 180° and find the flip on the compass again. The space between the two flips should be the required length of the rod.
****

On 10-22-00, I fully confirm that this lawn mower retrofitted with the GEET Multi-Fuel Processor is able to run very well with 75% of water and 25% of gasoline.
****

One must smooth off the end of the tee where the copper gasket meets to get a seal. This can probably be done with a file for most of the task. At auto parts stores, one can find valve-grinding compound in a little tube. It carborundum particles in a water base. Put a little between the two surfaces that must seal, spin them around a while and they will grind themselves to a fit. Clean off the grit with water. Hope this helps.
****

On one of my tests the bubbler jar was heated up and expanded and was being pushed up inside the juice can and falling back down again hitting the 2 screws that held the juice can to the muffler. Well since it was glass it broke and hot gas went all over the place BUT NO FIRE. I was lucky. I had used a wooden block in the bottom to prevent the jar from hitting the screws but it got so hot that the wooden block was smoking and charcoaling so I left it out thinking that it would not smoke up the jar.
MISTAKE Be careful. Now I am interested in temps and I will do some testing with an electric fry pan heating the bubbler to see what temp range is necessary and best for the vaporizing. I think I was heating it too hot.
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PANTONE'S "GEET" DEVICE is it REVOLUTION or energy DELUSION?
After several unsuccessful attempts by A.E.R.O. to purchase advertised plans for the much-touted "GEET" device, one of our researchers obtained a copy of Pantone's patent. We put together a device based on the information outlined in the patent information for the purpose of evaluation. A.E.R.O. researchers are convinced of the existence of "free energy" and other

advanced technology devices that could benefit the whole of Mankind; therefore, we approach such testing and evaluation of potential devices with great expectation. Sometimes we are elated with our findings, sometimes disappointed, and sometimes confused by the inventor's claims, but the approach is always the same: a positive rather than a "debunking" attitude, and we are continuously and tirelessly searching.

THE TEST

We attached our version of the "GEET" device to a small engine. Using a measured amount of 87-octane gasoline, we started the engine, making necessary adjustments to keep the engine running as long as possible. The first test was performed with a 4" rod ("Pantone reaction rod"). The engine ran fairly well, although somewhat roughly for eighteen minutes, one and sixty-seven one-hundredths seconds (00:18:01:67). There was virtually no detectable unwanted pollution from the exhaust, and the end of the exhaust pipe became warmer as the engine continued to run. During the operation, a slight magnetic disturbance was detected in the vicinity of the "reaction rod". When the engine stopped running, we observed an amount of waxy sediment floating on top of the remaining clear liquid, which we attributed to “additives”, contained in the gasoline, itself.

Test #2 was performed using the same amount of measured gasoline with an equal amount of water. The test was conducted in the same manner as the first, using the same 4" rod. The engine ran more smoothly and faster than observed in Test #1. The same observations during operation were performed with similar results, but the engine ran for only sixteen minutes, forty-three and eighty-nine one-hundredths seconds (00:16:43:89). Upon examination of the fuel container at test conclusion, we observed the same amount of waxy sediment and clear liquid floating on top of the entire measured amount of the added water. Conclusion: The device ran entirely on the combustible gasoline fumes minus the same "gasoline additives", but other than providing a better atmosphere for engine combustion, we could not conclude from this test that any water was consumed in the combustion process.

At the end of the two tests, we removed the 4" rod for examination. The rod did appear to have become magnetized with a somewhat weak field, but not enough to noticeably attract other metals. No other changes out of the ordinary were observed.

At this point, we substituted a 7 1/2" rod and proceeded to the next phase of testing.

Test # 3 was performed in the same manner as Test #2, using the 7 1/2" rod and the same measure of gasoline and water. The engine ran smoothly as observed in Test #2. The magnetic field in the vicinity of the rod appeared to be stronger, and the exhaust was noticeably cooler. The engine ran for a longer period than previously observed - nineteen minutes, twenty-seven and forty one-hundredths seconds (00:19:27:40). Upon examination of the fuel container, post-test observations revealed the same waxy sediment on clear liquid floating on the entire original measure of water. We could still not conclude from the test that any of the water was consumed during the combustion process.

Upon examination of the rod, the magnetization characteristics appeared stronger, but still not sufficient to attract other metals.

We feel that further testing is merited due to the observed test results, but some modifications to our assembly will be required before proceeding.

Based on this series of testing, we can only conclude, so far, that heated gasoline vapors aided by exhaust pressure from the engine causes the gasoline to burn cleaner and longer than in conventional methods, but we are faced with the problem of what to do with the accumulated sediment. We know that certain rocket engines operate on alcohol and water, so it is possible to break the water down into the separate combustible elements of hydrogen and oxygen under the right conditions. Perhaps we have failed to attain the proper accelerated heat from the engine's exhaust, which would permit ideal conditions to exist.

We welcome any comments or suggestions regarding testing or modifications to the device. We also welcome any offers to present a ready-built device for testing. We will make no test claims other than what we honestly observe.

We also invite Paul Pantone to provide us with a complete set of plans, so that we might further validate his claims. He should email Professor Shields at the following address: prof_shields@hotmail.com as soon as possible.
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SNG stands for Synthetic Natural Gas. The substance that is supposed to be created by GEET,
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This is why your systems are inefficient--heat alone does not cause cracking. A massive negative pressure is needed in the reactor. This is why less air in the reaction works better.
****

Tonight I tried something new as suggested by Mike Holler and it seemed to reduce my emissions further on my 4 HP lawnmower engine. I ran the GEET gas through another bubbler containing water, allowing it to pick up some water vapor prior to entering the intake. I have a feeling that this has increased my fuel efficiency as well since the water expands considerably. I plan to check this tomorrow. My prior fuel efficiency had been about 10.67 hours/gallon of gas at nearly full power, but under no load. We haven't had enough rain to make the grass grow here in New Jersey, so I haven't been able to run a real test on this yet. But, my latest run revealed very low emissions and tons of power.
****

We all have built our units to the plans with the12" outer pipe etc. Leaving cost out of the exercise and knowing some of the rods are less than 12" and hearing that nickel could assist - what is the idea of a 3/4" outer pipe a 1/2" inner tube and the unit 9" long all fabricated in "thin" stainless steel with smaller valves etc. I know the plans are in "principle" only but with the thinner walls etc & perhaps a deep threaded rod - Is this where the word vortex comes from ??? Should the exhaust be made to spiral around the inner tube ?? I know I'm jumping ahead & we have even answered those temperature/vacuum questions. Should we not restrict ourselves to standardizing the fuel we want to use.
****

we have used nickel and glass after they were taken out they displayed magnetic qualities, we cannot explain this but try it.
****

What I currently use is a piece of 1/16 wire in the end of the chamber that is closest to the exhaust manifold. It prevents the rod from going to far in the direction of the moving fuel. I did the same on the 8 hp generators that I retrofitted before I started this metro project. I used to have knobs on the rod, to keep it centered; now I don't. I still get the reaction.
****

Which is why I thought up an idea on creating a self-centering permanent magnet bearing. No electricity needed just a speaker magnet in the middle and rare earth magnets or equivalent strength on the outside in three or four

places. (Bruce, this is why I wanted some of those--where can I buy them and how much are they?) It is a setup that not only centers the shaft in the housing but also prevents end to end movement. This was originally a design I was going to use to make a permanent magnet motor but think of the other uses. Like a high speed supercharger for the GEET...
****

You don't need an air management valve. Start with a small engine like a mower. use the free geet plans BUT only use one valve to mix the air and geet gas before the engine intake. Leave the small carb where it is for throttle control and you will be a lot happier geetser. Use a jar bubbler so you can "see". It’s real pretty. It will be much cleaner exhaust.... Everything will get burned. Its fun to play with too.
****

You mentioned you had a problem with the inner chamber expanding and the seals vibrating loose? Well I currently have that problem now. The funny thing is that the north end stays tight but the south end loosens and the engine sucks to much air, then I have to shut it down. My idea to fix this is to weld bolt brackets (one on either end of the outer tube and one on each reducer) onto the reaction chamber . This should keep the seals tight and still allow for expansion of metal. I have not yet tried this yet due to lack of time but I will try it. I hope this idea may help you in some way
****